

Problem Statement: Shopping Cart System

Objective:

Develop a simple shopping cart system that allows users to add, remove, and view items in a shopping cart. The system should also calculate the total price of the items in the cart.

Requirements:

Item Class:

- Each item has an ID, name, and price.
- Implement methods to get item details.

Shopping Cart Class:

- Users can add items to the cart.
- Users can remove items from the cart.
- Users can view items in the cart.
- The cart can calculate and display the total price of all items.

Test Cases:

Item Class Tests:

- Test creating an item with valid details.
- Test getting item details.

Shopping Cart Class Tests:

- Test adding items to the cart.
- Test removing items from the cart.
- Test viewing items in the cart.
- Test calculating the total price of items in the cart.

Constraints:

- Use Java and JUnit for implementing and testing the system.
- Implement the system using Test-Driven Development (TDD) principles:
 - Write failing tests based on requirements.
 - Implement code to pass the tests.
 - Refactor and optimize the code as needed.